

Subject: bif401 (final term)

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- 1) Genomics, evolutionary studies and system biology are application of **bioinformatics**
- 2) Exact matching require which type of nucleotide-**same number of nucleotide**
- 3) Accurate solution of protein structure is one of the ____ **toughest problems**
- 4) **Tract back** is overlap matches can start from any position in scoring matrix
- 5) The matrix hasscores. **Positive and negative**
- 6) **MASCOT**____can search people mass finger printing and shotgun proteomics data set
- 7) Spectrometer measures the protein by_____ **Mass/charge**
- 8) Simulation of the folding process depends_____ **Energy function**
- 9) Information about **DNA/RNA** is available on gene bank
- 10)**TDP** is used to measure the molecules weight of intact proteins.
- 11)Coordinates of **alphacarbon** in the protein backbone can be used for structural visualization.
- 12)in structure visualization c-alpha atoms are traced to recreate a _____ **3Dproteinstructure**
- 13)in UPGMA distance is calculated btw _____ **twoclustersandbetweentwotrees**
- 14)pk value of Aspartic acid **3.9**
- 15)Largest amino acid **Tryptophan**
- 16)2nd STRUCTURE of proteins can be obtained by using_____ **ChouFasmanalgorithm**
- 17)NJ Algorithm strategy is used to predict ____ **RNA2` structure**
- 18)The loop of hair pin must be atleast____ in length **2aminoacids**
- 19)Fold recognition is also called **threading**
- 20)Ab initio method in contrast ,base their prediction on **lowenergymodel**.
- 21)Bioinformatics require.....smart mind, and connected to internet, **both**,
- 22)Sequence alignment tool is..... **PROSIGHTandMASCOT**

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- 23)MS1,MS2 provide us data identifying unknown.....**Protein's sequence**.
- 24)**MGF** files develop as an open standard for..... **proteomics data**.
- 25)**PDB** coordinates alpha carbon in protein back bone can be used for comparison
- 26)Alpha carbon atom can be obtained from.....**PDB**
- 27)in formation of 2nd structure of protein C & N can make.....**Hydrogen bonds**
- 28)information for protein folding is into its native structure is in.....
protein's amino acid sequence!
- 29)Energy **released** during bond formation.
- 30)hydrogen bonding occur in ...**secondary** ..Protein structure
- 31)Bond between C and H in alpha helices____**Hydrogen bonding**
- 32)Amino acid having 3 codons____**isoleucine**
- 33)MALDI typically adds a to protein or peptide. **Proton**
- 34)low quality match gets____**low score**
- 35)____ are sequences of amino acids produced during MS2. **PST**
(peptide sequence tags)
- 36)MS2 data can be used to extract..... **peptide sequence tags**.
- 37)The alpha helices propensity should be more than____**(1.0)**
- 38)How many types are of protein sequencing techniques__ **(2)**
- 39)Blast can find sequences of ____**nucleotide amino acid**
- 40)If a protein sequence of 26 amino acids is fragmented at 11 amino acid**C11, Z25**
- 41)To find out unknown sequence of nucleotide we use . **NGS or Mass spectrometry**
- 42)How many types of peptide May be injected in mass spectrometry chamber.. .
three hundred thousand to four hundred thousand
- 43).**MASCOT**.can search peptide mass finger printing and shotgun proteomics dataset.
- 44)Stabilizing and destabilizing energy give us **quality of 2nd structure** .
- 45)The Fasman algorithm is based on **statistical occurrence** of Amino Acids in known structures.

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46) DP recombinant the nucleotide recombination through process of

Traceback

47) **MALDI**.....add proton to a protein and a peptide..

48) Positively charged amino acids are..... **3**

49) Aromatic amino acid include **phenylalanine, tyrosine tryptophan**

50) Homology modelling fail to predict **quality structure**

51) How many forces are involved in protein folding. **4**

52) Pairwise alignments tells the similarity between sequence... **by maximizing the matches.**

53) Dot plots employ dot matrix with two sequence plotted represents.....**pairwise alignment and comparison.**

54) Differentiate b/w DNA and RNA sequence..... **RNA has more variety of sequence**

55) Cell molecule are produced after transformation of **DNA to protein**

56) Dot plot cannot deal with..... **Insertion, deletion and gaps.**

57) Exons are may be more **conserved.**

58) In dynamic algorithm we can do comparison of more sequence. **Three**

59) Mass/Charge ratio helps calculate the mass of the **Protein**

60) In scale tree branches lengths are equal to themagnitude of change in the **nodes.**

61) How many types of peptides mix in MS chamber..... **300,000 – 400,000**

62) mRNA is a structure. **Planer**

63) When unpaired bases of 2' stricture join to form..... **3' structure**

64) N Jackobson use to pridict **2' structure.**

65) Aromatic amino acid **Tyrosine**

66) Positive charge amino acids..... **Lysine, arginine, histidin**

67) MS begins with the measure of **intact protein**

68) Amount of amino acids in alpha helix..... **(4)**

69) Intact mass of protein can be found by..... **MS1**

70) MSA can be done by..... **CLUSTAL**

71) Which is used as sequence alignment tool..... **PROSIT**

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- 72)MS1 and MS2 help to identify (**PST**)
- 73)MGF is used for **proteomics**
- 74)X,Y,Z positions of alpha carbon available online on (**PDB**)
- 75)Method to determine RNA structure NMR, X-Ray crystallography**
- 76)CLUSTLA runs..... Slow accurate/ fast appropriate**
- 77)Which give info about precursor protein..... **PSTS**
- 78)Raw data files format can be converted to **Open format**
- 79)Which of the following is cartoonic figure..... Ribbon diagram**
- 80)In CATH protein recognize according to their..... structural similarity**
- 81)Method for obtaining 1^o structure..... (Edman Degradation Tandem Mass Spectrometry).**

Q. As MS chamber is only be used for ionization then, how can we make possible further fragmentation of protein in it?

ANS: For this sac there will be an addition of fragmentation instrument in MS chamber, which can perform this job nicely.

Q. Name the techniques used for protein fragmentation?

- ANS:1. Electron capture dissociation
2. Electron transfer dissociation
3. Collision induced dissociation

Q. Define PSTs? No option

Ans:peptide sequence tags (PSTs) are actually small amino acid sequence of peptides producing during MS2.

Q. what are three major types of scoring scheme which obtain from protein sequence data base search algorithm?

Ans: 1) MW match score 2) PST match score 3) invitro and in silico match score

Q. Name two types of open formats?

Ans: (1) mzXML (2) MGF

Q. Name some companies that are making mass spectrometer in market?

Ans: Agilent ----- Bruker ----- ABI ----- Micromass ----- Waters ----- Shimadzu

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Q. How many types of masses are there? Mention their names.

Ans: 1) Nominal Mass 2) Monoisotopic Mass 3) Average Mass

Q. Differentiate between polar and non polar amino acids?

Ans: Non polar amino acids:

Non polar amino acids are hydrophobic in nature. They are mostly found inside of folded protein. H and C introduce very little dipole movement. e.g Phe, Ala, Leu, Val etc.

Polar amino acids:

These amino acids are polar but are not charged. Prefer to reside or interact to aqueous environment. Mostly found at surface of folded protein. E.g Cys, Gly, Ser, Thr, Tyr etc.

Q. What is meant by “PK” value of an amino acid?

Ans: PK is the value for an amino acid is that PH at which exactly half of amino acids are charged and half are not charged.

Q. what are the key factors participating in protein folding?

Ans: solubility and reactivity are the key factors that are participating in protein folding.

Q. Mention some unique properties of amino acids? No option

Ans: amino acids has unique set of properties such as charge state, polarity and hydrophobicity.

Q. write two methods which we use to obtain 1st structure? No option

Ans: 1. Edman degradation
2. tandem mass spectrometry

Q. briefly describe any three properties of Loops? NO option

Ans: • Loops connect helices and sheets • Loops vary in length and 3-D configurations • Loops are mostly located on the surface of proteins • Loops are more “acceptable” of mutations • Loops are flexible & can adopt multiple conformations • Loops tend to have charged and polar amino acids • Loops are frequently components of active sites

Q. Define angstrom. No options

Ans: Angstrom is a unit of length equal to 10^{-10} m (one tenbillionth of a meter) or 0.1nm .

Q. Name some atomic representation techniques which are helpful in protein structure visualization? No options

Ans: These include CPK, Ribbon and Balls & Stick diagrams.

Q. Name any two Techniques for structure determination. No options

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Ans: 1. X-Ray Crystallography
2. Nuclear Magnetic Resonance (NMR) Spectroscopy

Q. What are the other possible structures which beta strands can give rise to?

ANS: 1. Beta Sheet 2. Beta Barrel 3. Beta Sandwiches

Q.. How many types of Domains are there?

ANS: 1). Alpha Domains 2). Beta Domains
3). Alpha/Beta Domains 4). Alpha + Beta Domains
5). Alpha & Beta Multi-Domains 6). Membrane & cell-surface proteins

Q..Name the Algorithms use for Structure Classification? No option

Ans: 1. CATH 2. CSOP 3. FSSP 4. Pclass

Q.What is the abbreviation of I-TASSER?

Ans: Iterative threading assembly refinement (I-TASSER)

Q. How many Strategies are there for Structural Modelling?

Ans: There are three basic Strategies,

1. Homology Modelling
2. Fold Recognition
3. Ab Initio Modelling

Q. How many basic types of Alignments, mention their names ?

Ans: there are two types of alignment,

Global Alignment (Needle Wunsch)
Local Alignment (Smith Waterman)

Q.How many types of Phylogenetic Trees are there?

Ans: There are two types of phylogenetic trees:

- Rooted Trees
- Unrooted Trees

Q. ExPASy stand for ?

Ans: ExPASy is one of the best collection of data and tools for proteomics. It stand for "Expert protein analysis system"

Q. differentiate between bottom up and top down proteomics?

Ans: There are two major types of MS-based proteomics e.g mass spectrometry proteomics,

"Bottom up proteomics deals with peptides".

"Top down proteomics can handle whole proteins".

Bottom up proteomics measures the peptide masses produced after protein enzymatic

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digestion. And Top down proteomics measures the intact proteins followed by peptides after fragmentation.

Q. how loops are found after the finding of alpha and beta sheets?

Ans: After computing the propensity of alpha helices and beta sheets, we need to settle for loops.

- Let's see how can we find out the loops using Chou Fasman Algorithm.
- For any jth residue in sequence, we calculate $f(\text{Total}) = f(j) f(j+1) f(j+2) f(j+3)$ (tetra peptide).
- If,
- $f(\text{Total}) > 0.000075$.
- The average value for $P(\text{turn}) > 1.00$ in the tetra peptide.
- The averages for the tetra peptide are such $P(\text{a-helix}) < P(\text{turn}) > P(\text{b-sheet})$,
- It is a Turn!

Q. . How protein structure identified?

Ans: If another protein which has a similar sequence also has its structure known, the structure of an unknown protein can be predicted based on that similar protein.

So, it is then possible to identify unknown protein structures by just examining the homologous protein sequences. Good sequence alignment and identity ensures that homology modelling will give accurate results

Thus, Homology modeling is used to predict structures of proteins having high sequence similarity with other proteins with known structures:

Q. What happens when pH is < or > than pK?

Ans: If $\text{pH} < \text{pK}$ for an amino acid the amine side chain gain a proton (H^+) and become a positively charged, hence basic.

If $\text{pH} > \text{pK}$ for an amino acid, the carboxyl side chains loses a proton (H^+) and become negatively charged, hence acidic.

Q. What are the properties and role of amino acids?

Ans: Properties:

Amino acids have several properties such as charge state, polarity and hydrophobicity Amino acids not only have physical and chemical properties, but also structural properties

Role:

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These structural properties are equally important in giving rise to protein structures. Since some amino acids are hydrophobic, they may be employed in forming a stable core in a protein.

- Also, chemically inactive amino acids reduce chances of destabilizing reactions in core.

Q. Why folding is important?

Ans: Proteins fold spontaneously. Proteins fold to achieve thermodynamic stability. Proteins fold to organize themselves for performing functions in cells.

Q. What is ab initio modelling? Describe the limitations of this model?

Ans: Ab initio methods have Anfinsen's thermodynamic hypothesis at the center.

- These methods attempt to identify the structure with minimum free energy.

Limitation:

- Computationally expensive
- Suitable for proteins with less than 100 residues

Q. Write down the terms of PDB file format.

Ans: **HEADER:** contains the brief description of the structure, the date and PDB ID code.

TITLE: title of the structure.

SOURCE: identifies which organism the structure came from.

COMPND: brief detail of the structure.

REVDAT: the date of the last revision.

JRNL: one or more literature reference that describes the structure.

KEYWDS: list the set of useful words/phrases that describe the structure.

AUTHOR: the scientist depositing the structure.

REMARKS: detail of the experimental method used to determine the structure are contained in this subsection.

Q. Define ionization in terms of proteins?

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Ans: ionization of protein means to ionize a protein molecule to make it charged by removing an electron from it. Protein ionization is the first step in MS-based proteomics protocols

- Ionization process involves loading of a proton onto a protein or the removal of a proton from the protein

- Ionization results in an increase or decrease of the protein/peptide mass

Two primary techniques used for ionization of protein,

1. Electrospray ionization (ESI)

2. Matrix - assisted laser desorption/ionization (MALDI)

Q. Briefly describe Choufasman algorithm?

Ans: **Chou – Fasman method:**

It is a technique for the prediction of secondary structures in proteins i.e.

Alpha Helices, Beta Sheets and Turns is Chou – Fasman technique. The method is based on analyses of the relative frequencies of each amino acid in alpha helices, beta sheets, and turns based on known protein structures solved with X-ray crystallography. From these frequencies a set of probability parameters (in our handouts, it is propensity table) were derived

- **For the appearance** of each amino acid in each secondary structure type.

- **To predict the probability** that a given sequence of amino acids would form a helix, a beta strand, or a turn in a protein.

Q. How can alpha helix be expressed?

Ans: **Chou-fasman algorithm (alpha helix):**

- For Alpha Helices, 4 contiguous amino acids are required.
- Their Alpha-Helix propensity should be more than 1.0
- Once this propensity falls below 1.0, Alpha-Helix stops.

Q. Describe scores in silico fragmentation?

Ans: **Silico fragmentation scoring:**

- Count the matches between in silico and in vitro peaks.
- Give an equivalent score to the candidate protein.
- Weigh each of the aforementioned match by the mass error.
- Accumulate the score

Q. What are hydrophobic amino acids?

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Ans:Hydrophobic Amino Acids:

Since H and C introduce very little dipole moments in hydrophobic amino acids, these amino acids are non polar. Hydrophobic amino acids are mostly found at the inside of folded proteins.

Q. Explain summary of alpha helix and beta sheets?

Ans:Alpha Helices are formed by hydrogen bonding (O-H) between C_i and N_{i+4} atoms in the protein backbone. X-Ray Crystallography of protein shows that Hydrogen atoms of N-Term come together with Oxygen atoms of C-term amino acid at 4th neighboring position to make Hydrogen bonds!

Beta Sheets are another common secondary structure

- They are constituted by several Beta Strands which come together
- 5 to 10 residues are needed to make a Beta Strand, typically

Q. what is 3D_1D bowie algorithm?

Ans: Proposed by Bowie et al in 1991

- Converts 3D structure into a 1-D string profile for each structure in the fold library
- Align the target sequence to these profiles 3D-1D methods convert structure and environment information into "profiles"
- Score for each amino acid is computed for each profile

Q. public database that commonly used for protein?

Ans: Examples include Uniprot, Swissprot amongst several others.

Write steps of protein sequence identification? OR Enlist protein sequence method.

Ans: Mass spectrometers are used to measure the molecular weight of proteins and peptides.

Following steps involve in protein sequence identification,

1. complex protein
2. separation
3. ionization by mass spectrometry
4. fragmentation MS2
5. mass spectra

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6. EST Determination
7. filter protein database
8. in silico fragmentation of candidate protein
9. matching of experimental and insilco peak list
10. post translational modification
11. protein score.

Define Peptide mass finger printing nd shotgun?

Ans: Shotgun proteomics digests the entire protein mix 1st followed by peptide analysis & protein databasesearch

- Peptide mass fingerprinting involves proteinseparation followed by a single protein's peptideanalysis.

Q.Principal of dynamic programming?

"Dynamic programming is an algorithm technique used commonly in sequence analysis".

Select a nucleotide from RNA sequence

AUGCUGUCAUCG

- List all possible complementary positions for selectednucleotide in the complete sequence

AUGCUGUCAUCG

AUGCUGUCAUCG

AUGCUGUCAUCG

AUGCUGUCAUCG and so on...

DP exhaustively makes all possible nucleotidecombinations

- DP then recombines such combinations in a processcalled "Traceback" to ensure that the highest coupled2' structure is reported.

Q. write types of RNA ?

ans:**Coding RNAs**

- Coding RNAs as is obvious from their name , code for proetein
- **Non-coding RNAs**
- Non-coding RNAs regulate/assist in the process of translation.Q

Other types of RNA are,

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1= m RNA

2=t RNA

3= r RNA

4= miRNA

5= si RNA

Q.: Explain different databases and their specificity.

Ans: There are two basic types of data bases,

1. Gene Bank:

It is an online public sequence data base. In Gene Bank we go to online NCBI Gene Bank to find

sequence of DNA and RNA using following Website, www.ncbi.nlm.nih.gov/genbank
Gene Bank can be research by typing,

1. Sequence name
2. Species
3. ID
4. Locus
5. Author
6. Journal
7. Accession number
8. Name

2.Uniport:

It is also an online public data base. By using Uniport we can find sequence of proteins.
If

we want to find a protein sequence we must go an online Website given bellow,
www.uniprot.org

In home page there is a box named “**Swiss Prot**” which contains human curated protein information,

molecular mass, observed and predicted modifications etc.

uniport can be research by typing.

1. Amino acid
2. Name
3. ID
4. Sequence

Further Data Bases and their specificity is mention bellow,

Taxonomic Database For classification

Genomic Database For Gene level information

Q. Define Top down proteomics?

Ans: “Top down proteomics can handle whole proteins”.

Top down proteomics measures the intact proteins followed by peptides after fragmentation.

Q. Need for Choufasman algorithm?

Ans: Choufasman algorithm is a propensity base method. The method is based on analyses of the relative frequencies of each amino acid in alpha helices, beta

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sheets, and turns based on known protein structures solved with X-ray crystallography. From these frequencies a set of probability parameter were derived.

Q. Write down an clustering

Substitution	ACGA
s	→□AGGA

objective base method and approach?

Ans: UPGMA is a clustering approach and least square distance is an objective base method.

Q. Differentiate between finger printing and short gun proteomics?

Ans: Shotgun proteomics digests the entire protein mix 1st followed by peptide analysis & protein database search

- Peptide mass fingerprinting involves protein separation followed by a single protein's peptide analysis.

Q. what are substitutions and indels?

Ans: Indels:

Removal and addition of amino acids in proteins and nucleotides in DNA, RNA by using Gaps named as Indels.

Substitution:

Replacement of a single or a specific pair of amino acid or nucleotide with other on in a sequence is called substitution.

Q. How can psuedonots detected in the prediction of RNA?

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Ans: Tertiary or 3' structure of RNA may form pseudoknots to detect the pseudoknots in RNA

structure we need "circular plot" which is a graphical approach.

Q. Formation of protein sequence and limitation?

Ans: There are two basic techniques for sequence of a protein,

1. Edman degradation

2. Mass spectrometry

Edman degradation used to sequence the smaller length protein, about 40 – 50 amino acids.

Large protein sequence is carried out by mass spectrometry.

*Limitations nimili,

Q. Uniprot and ESEMBL or EMBL?

Ans> **Uniprot** is a public data base which is used to find the sequence of protein.

ESEMBL is genome search engine which is used to search the genome of every recorded

species. OR it might be EMBL so,

Input to FASTA search can be in

* **EMBL***, = European molecular biology laboratory (EMBL)

Q. Mass spectrometer?

Ans> Mass spectrometer is used to measure mass/charge ratio of ionized proteins and peptides.

Data output from the MS comprises of m/z ratios and intensities of each molecule that is measured.

Function of Tandem MS:

Tandem MS helps in measurement of mass to the fragments as well. This process provides another step in further scoring and ranking and Protein identification thus becomes easier.

Q. Define Orthology?

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Ans> it is a homologous gene present in two organisms that encode protein with same function. It is evolved by direct vertical descent.

Q. What does role of 5' cap and 3' poly A tail?

Ans: the 5' end of molecule capped with 7 methyl guanosine tri phosphate and 3' end poly A tail

cap, both caps play major role to translate mRNA to ribosome, also protect mRNA.

Q.: Explain applications of bioinformatics.

There are some major applications of bioinformatics:

In this field bioinformatics help us in gene finding, in assemblage of genes and forming of databases.

2. Proteomics:

It helps us to decoding protein sequences for better understanding about protein structure,

protein to protein relationship, post translational changes and in generating databases for their

sequence and structure.

3. Evolutionary study:

It can make phylogenetic trees to find evolutionary relationship between species, also show

ancestry between them,

4. System biology:

Bioinformatics also helps us in regulatory mechanism in genes and proteins. So that we can

better understand about regulators and treat them by evaluate drugs.

Bioinformatics also help us in further many aspects like,

Drug development

Waste cleanup

Gene therapy

Preventative medicine etc.

Q=write types of phylogenetic trees ?no option

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Rooted trees:

- Each node with descendants represents the inferred most recent common ancestor of the descendants.
- The edge lengths in some trees may be interpreted as time estimates.
- Rooted trees can show temporal evolutionary direction.
- Expensive.

Unrooted:

- Only the relatedness of the leaf nodes.
- Do not require the ancestral root to be known or inferred.
- Less expensive.

Uses:

In bioinformatics, such as

- Rooted and unrooted trees can be used to show phylogenetic relationships between sequences.

Q. define open reading frame?

In molecular genetics, an **open reading frame (ORF)** is the part of a **reading frame** that has the potential to code for a protein or peptide. An **ORF** is a continuous stretch of codons that do not contain a stop codon (usually UAA, UAG or UGA).

Q=what is FASTA? Why we use it?

- FASTA can perform quick comparison of protein and nucleotide sequence
- Can also perform genome and proteome similarities search
- Its also available online like BLAST.

FASTA can search sequence databases and identify unknown sequences by comparing them to the known sequence databases. This can help obtain information on the parent organism, function and evolutionary history

Q.What is BLAST ? NO OPTION

➤ BLAST finds regions of similarity between biological sequences. The program compares nucleotide or protein sequences to sequence databases and calculates the statistical significance. In bioinformatics, **BLAST** for **B**asic **L**ocal **A**lignment **S**earch **T**ool is an algorithm for comparing primary biological sequence information, such as the amino acid sequences of proteins or the nucleotides of DNA sequences. A BLAST search enables a researcher to compare a query sequence with a library or database of sequences, and identify library sequences that resemble the query sequence above a certain threshold.

➤ Different types of BLASTs are available according to the query sequences. For example, following the discovery of a previously unknown gene in the mouse, a scientist will typically perform a BLAST search of the human genome to see if humans carry a similar gene; BLAST will identify sequences in the human genome that resemble the mouse gene based on similarity of sequence.

Q. What is the difference between basic and acidic amino acids?

Ans: the difference is in their side chains, the acidic ones have acidic side chain while basic ones have basic side chain at neutral pH.

- Arginine, lysine and histidine are the basic ones.
- Aspartate and glutamate are the acidic ones.

Q. What is scope of bioinformatics?

Ans. Bioinformatics primarily deals with digitalized biological information as well as data reported from biology experiments. Computational methods, data processing techniques and algorithms are used in addressing the following issues,

Use of computational algorithms and techniques for:

1. Storage,
2. Organization,

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3. Analysis, and
4. Representation of biological information

How junctions are formed?

Ans: junctions are formed when two or more double stranded regions of RNA converging to form a closed structure.

Q. Name of famous protein data base?

Ans: SWISS PROT and UNI PROT

Q. Four objectives of comparing sequence?

- Ans:**
1. Similarities among sequences
 2. Differences b/w sequences
 3. Evolutionary history
 4. Predict the function of molecules

Q. How bulges are formed?

Ans: in RNA 2' structure bulges are formed when double stranded region cannot form base pairs perfectly.

Q. how biological simulation help community?

Ans: it helps community by predicting disease, progression and outcomes.

Q. what is role of EXPASY?

Ans: it provides access to a variety of online data bases and tools depending upon our requirements. We can find sequence information by it.

Q. Define PSTs? No option

Ans: peptide sequence tags (PSTs) are actually small amino acid sequence of peptides producing during MS2.

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Q. how loops are found after the finding of alpha and beta sheets?

Ans: After computing the propensity of alpha helices and beta sheets, we need to settle for loops.

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- Let's see how can we find out the loops using Chou Fasman Algorithm.
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Q. . How protein structure identified?

Ans: If another protein which has a similar sequence also has its structure known, the

structure of an unknown protein can be predicted based on that similar protein.

So, it is then possible to identify unknown protein structures by just examining the homologous protein sequences. Good sequence alignment and identity ensures that homology modelling will give accurate results

Thus, Homology modeling is used to predict structures of proteins having high sequence similarity with other proteins with known structures:

Protein structure prediction:

There are three different strategies for structure prediction

1. Homology Modelling
2. Threading/Fold Recognition
3. Ab Initio Modelling.

Q. What are the properties and role of amino acids?

Ans: Properties:

Amino acids have several properties such as charge state, polarity and hydrophobicity Amino acids not only have physical and chemical properties, but also structural properties

Role:

These structural properties are equally important in

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giving rise to protein structures Since some amino acids are hydrophobic, they may be employed in forming a stable core in a protein

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Ans: **Chou – Fasman method:**

It is a technique for the prediction of secondary structures in proteins i.e.

Alpha Helices, Beta Sheets and Turns is Chou – Fasman technique. The method is based on analyses of the relative frequencies of each amino acid in alpha helices, beta sheets, and turns based on known protein structures solved with X-ray crystallography. From these frequencies a set of probability parameters (in our handouts, it is propensity table) were derived

- **For the appearance** of each amino acid in each secondary structure type.

- **To predict the probability** that a given sequence of amino acids would form a helix, a beta strand, or a turn in a protein.

Q. How can alpha helix expressed?

Ans: Chou-fasman algorithm (alpha helix):

- For Alpha Helices, 4 contiguous amino acids are required.
- Their Alpha-Helix propensity should be more than 1.0
- Once this propensity falls below 1.0, Alpha-Helix stops.

Q. Describe scores in silico fragmentation?

Ans: Silico fragmentation scoring:

- Count the matches between in silico and in vitro peaks.
- Give an equivalent score to the candidate protein.
- Weigh each of the aforementioned match by the mass error.
- Accumulate the score

Q. what is 3D_1D bowiealgorithm?

Ans: Proposed by Bowie et al in 1991

- Converts 3D structure into a 1-D string profile for each structure in the fold library
- Align the target sequence to these profiles 3D-1D methods convert structure and environment information into “profiles”
- Score for each amino acid is computed for each profile

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Define Peptide mass finger printing nd shotgun?

Ans: Shotgun proteomics digests the entire protein mix 1st followed by peptide analysis & protein database search

- Peptide mass fingerprinting involves protein separation followed by a single protein's peptide analysis.

Write steps of protein sequence identification? OR Enlist protein sequence method.

Ans: Mass spectrometers are used to measure the molecular weight of proteins and peptides.

Following steps involve in protein sequence identification,

1. complex protein
2. separation
3. ionization by mass spectrometry
4. fragmentation MS2
5. mass spectra
6. EST Determination
7. filter protein database
8. insilico fragmentation of candidate protein
9. matching of experimental and insilco peak list
10. post translational modification
11. protein score.

Q. write types of RNA ?

ans: **Coding RNAs**

- Coding RNAs as is obvious from their name , code for proetein

• Non-coding RNAs

- Non-coding RNAs regulate/assist in the process of translation.Q

Other types of RNA are,

1= m RNA

2=t RNA

3= r RNA

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4= miRNA

5= si RNA

Advantage and disadvantage of Ab initio:

Advantages:

- Ab Initio methods can fold any target sequence using only physical atomic properties.
- Predictions are mostly accurate and correctly describe the natural folding process.

Disadvantages:

- Ab initio methods are the very difficult to design (energy function).
- These methods are slow due to the huge possibilities

PAM matrix

- PAM means "Point Accepted Mutations"
- Point accepted mutation is a substitution of one amino acid by another such that the protein functions stays conserved.
- PAM unit is a time in which about 1% of amino acids in a sequence undergo accepted mutations
- PAM matrices are scoring matrices that are useful in computing sequence alignment scores.

STEP TO COMPUTE PAM MATRICES

1. Align the protein sequence which are 1-PAM Unit diverge.
2. Let A_{ij} be the number of times A_i is substituted by A_j .
3. Compute the frequency f_i of amino acid A_i .

Then, $PAM1 = p_{ij} = \frac{A_{ij}}{\sum_k A_{ik}}$

PAM 'n' = (PAM1)

Steps present in flow chart of homology modeling: - (10 marks)

1. Template recognition and initial alignment.
2. Alignment correction.
3. Backbone generation.
4. Loop modeling.

5. Side-chain modeling.
6. Model optimization.
7. Model validation

How homology modeling is used for knowing the sequence of unknown protein sequence: (10 marks)

If another protein which has a similar sequence also has its structure known, the structure of an unknown protein can be predicted based on that similar protein.

So, it is then possible to identify unknown protein structures by just examining the homologous protein sequences. Good sequence alignment and identity ensures that homology modelling will give accurate results

Thus, Homology modeling is used to predict structures of proteins having high sequence similarity with other proteins with known structures:

How we can get scoring in BUP proteomic

There are two approaches to perform BUP:

- Peptide Mass Fingerprinting.
- Shotgun Proteomics

Score in Protein Fragmentation:-

If we can:

- Measure the mass of fragments using MS.
- Calculate the Protein Fragmentation Techniques theoretical mass of the fragments.

Then, we can award score on the basis of the similarity of experimental and theoretical mass

Protein Fragmentation techniques:

- Electron Capture Dissociation (ECD).
- Electron Transfer Dissociation (ETD).
- Collision Induced Dissociated (CID).

Types of secondary structures of RNA:

1. Single stranded:

- 3'end may fold on to the 5' end.

2. Helices:

- Double stranded RNA helix of stacked base pairs

3. Hairpin loop:

- The loop of the hairpin must at least four bases long to avoid steric hindrance with base-pairing in the stem part of the structure

4. Bulge Loops:

- Bulges, are formed when a double-stranded region cannot form base pairs perfectly

5. Interior loop:

- Interior loops are formed by an asymmetric number of unpaired bases on each side of the loop.

6. Junctions or intersections:

- Junctions include two or more double-stranded regions converging to form a closed structure.
- The unpaired bases appear as a bulge.

Q. How to build scoring matrices?

Ans: To build the Scoring Matrices we analyze the amino acids and nucleotides which are substituted in single gene and protein sequence.

Scoring Matrices have both values +ve and -ve. Positive value for matches and negative value

for mismatches. Different type of scoring matrices can be developed based on underlying strategy.

1. Each amino acid have different property.
2. Each amino acid have different frequency.
3. When we compare the sequences they match and mismatch according to their frequency.

Q. Optimal energy function in structural prediction?

Ans: Energy based methods involve evaluating the free energy structures. To compute the RNA

sequence for 1' or 2' optimal structure prediction we use Zuker's Algorithm.

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RNA Secondary Structure Prediction.

Zuker's Algorithm helps us to compute the stabilizing energies (-ve) and also destabilizing

energies (+ve values). And also compute the sum of +ve and -ve energies. All possible 2' structures are generated. The best 2' structure is selected.

Q. Home message of this course?

Ans: 1. Introduce the classical algorithms in bioinformatics

2. Link them to latest developments in the field

3. Evaluate the future applications

Q. Diff. b/w MS and Raw file data?

Ans: **Mass spectrometer** (MS) outputs data with mass/charge ratios & respective ion intensities.

RAW file is a format in which an instrument outputs data in binary form. Several software exist for converting RAW file formats into open software formats. Each open format has its own unique advantages. mzXML and MGF formats are most frequently used

Q. Why we need make protein structure when we have X-Ray and NMR?

Ans: There are thousand of sequences and lesser structures are available. Reason is that some proteins cannot be crystalized or it may not be soluble so we have need to make protein structure althou we have x- ray crystallography.

Q. Amino acid with 3 codon?

Ans: lysine has 3 codon. (Lie = AUU, AUG, AUA)

Q. How + and – energies released in molecule?

Ans: The stabilizing energy associated with stacking base pairs in a double-stranded region is (-ve) energy.

• The destabilizing influence of unpaired regions is (+ve) energy.

Q. How back bone of protein form?

Ans: C termini in a protein backbone C-Alphas can be used to construct the backbone of a

protein towards its visualization. Proteins have Carbon and Nitrogen in their backbone

Q. which factor consider during N-J trace back?

Ans: 'Trace back' strategy is used to recover the optimal structure, there can be multiple trace backs. Each trace back can be used to construct an RNA secondary structure. NJ Select the trace back with the highest number of coupled nucleotides.

Q. Top down proteomics?

Ans: for a direct measurement of protein mass

- Solution: TDP

1. Samples containing the protein mixture from cells or tissue are obtained
2. The entire protein mix is analyzed for protein masses (MS1)
3. The mass list thus obtained has masses of all intact proteins
4. Note that proteins typically carry post-translational modifications and TDP caters for their mass measurement as well
5. After MS1, one protein is selected at a time and fragmented to obtain its peptides (MS2)
6. Several protein fragmentation techniques exist for fragmenting proteins
7. Fragmentation products i.e. peptides are measured for their mass (MS2)
8. The process is repeated (MS'n')

Conclusions:

- TDP measures the mass of intact proteins
- Mass of post translational modifications can be easily measured.

Q. Formula of scoring matrices?

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Ans: $p_{ij} = \frac{A_{ij}}{\sum_k A_{ik}}$

Q. What are forces used in amino acids to fold a protein?

Ans: Electrostatic interactions

- van der Waals interactions
- Hydrogen bonds
- Hydrophobic interactions

Q. How OH of RNA make it less stable than DNA?

Ans: The two hydroxyl groups make RNA less stable than DNA because it is more prone to hydrolysis. Hence, RNA has a short life span and is more prone to degradation.

Ab initio modeling:

Advantages:

Ab Initio methods can fold any target sequence using only physical atomic properties so atomic properties, electrostatic forces and molecular interaction all these included in it. So in this way the structure predicted by it are mostly accurate and correctly describe the natural folding process.

As for structure prediction in the case of threading and homology we need to compare our unknown sequence with the known sequence of PDB, so if there is a sequence which is not match with any data base Ab initio modeling is a best way to predict its structure.

Disadvantages:

In Ab initio modeling if we have a sequence we will form multiple structure from it. Then we compute energies for all these possible structures and select one with lowest energy. In some cases our selected structure does not have biological plausibility, means it could have some problems of misfolding or in its membrane.

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Ab initio methods are the very difficult to design. Because it require energy function it is very difficult to find.

As to compute the energy while making structure we have to find the energy of whole sequence, and if there is a very long protein sequence it will be too much cumbersome to compute the overall energy. So in this regard it is very expensive method to deal all such circumstances.

These methods are slow due to the huge possibilities
It can only deal with shorte